

DATA STRUCTURE

LAB PROGRAMS

5. Write a C program to find Factorial of a given number using Recursion

```
#include <stdio.h>

long int factorial(int n)
{
    if(n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}

int main()
{
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    printf("Factorial of %d = %ld", n, factorial(n));
    return 0;
}
```

Output

```
Enter a number: 5
```

```
Factorial of 5 = 120
```

```
=== Code Execution Successful ===
```